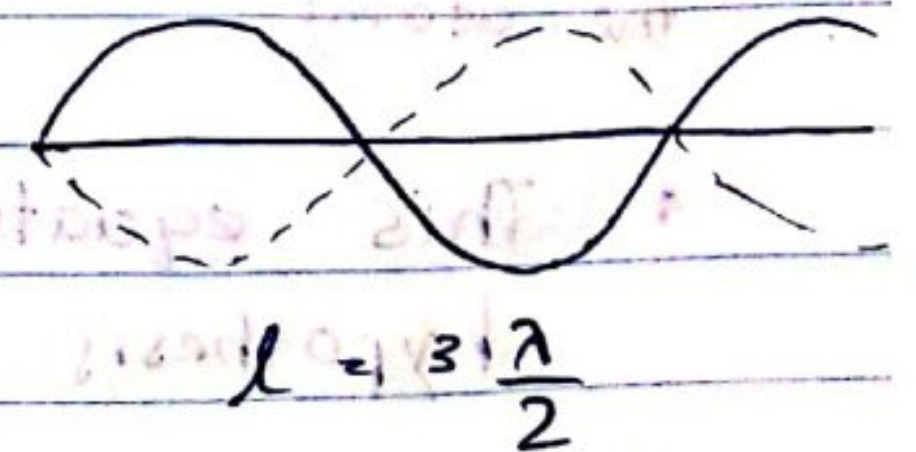
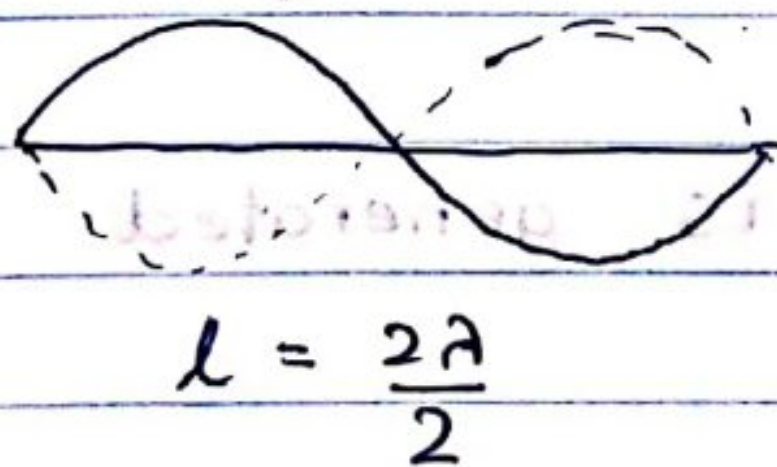
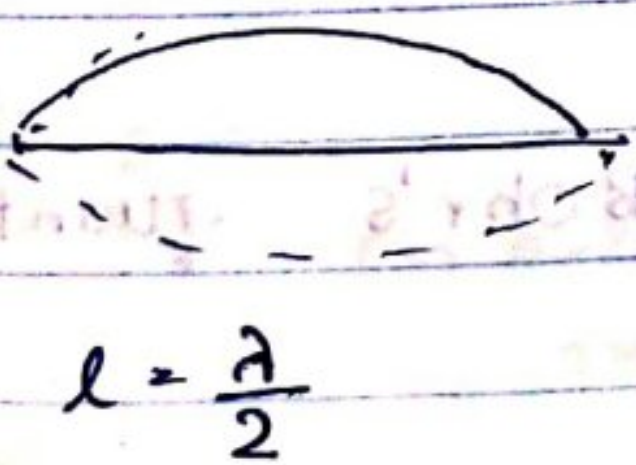
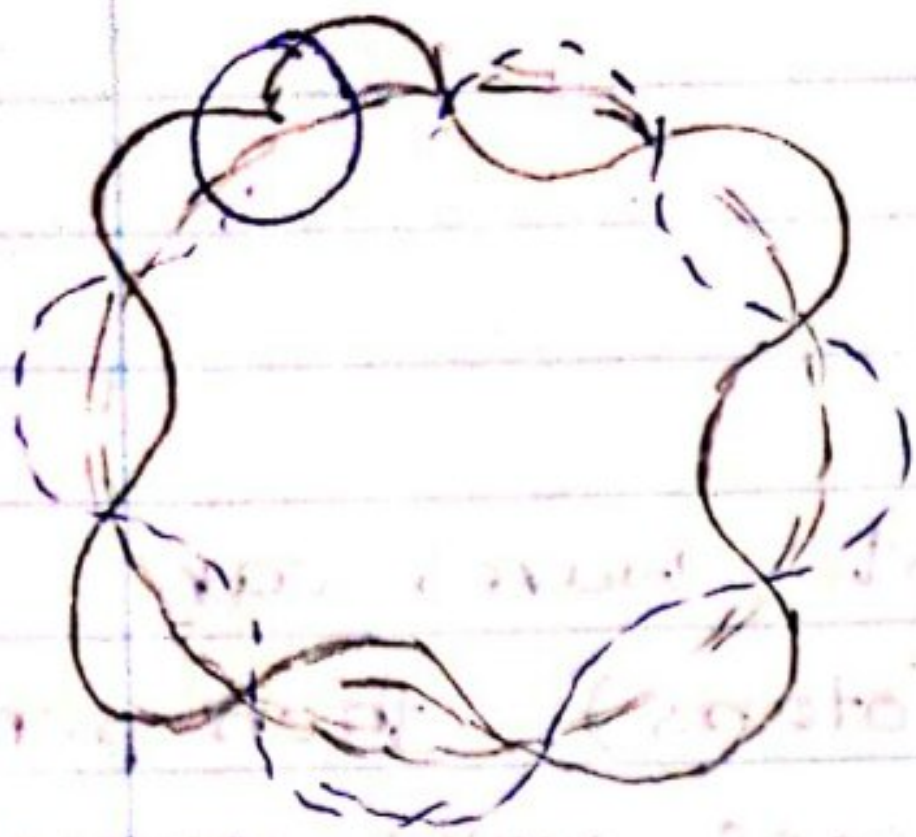


Bohr Model of the atom (The Dual Nature of the electron)

- * De Broglie reasoned that if light waves can behave like a stream particles (photons), then perhaps particles such as electrons can possess wave properties.
- * According to De Broglie, an electron bound to the nucleus behave like a standing wave.
- * The waves are described as standing (stationary) because they do not travel along the string.
- * Some points on the string, called nodes (do not move at all; that is, the amplitude of the wave of these points is zero)
- * There is a node at each end and there may be nodes between the ends.
- * The greater the frequency of vibration, the shorter the wavelength of the standing wave and the greater the number of nodes.

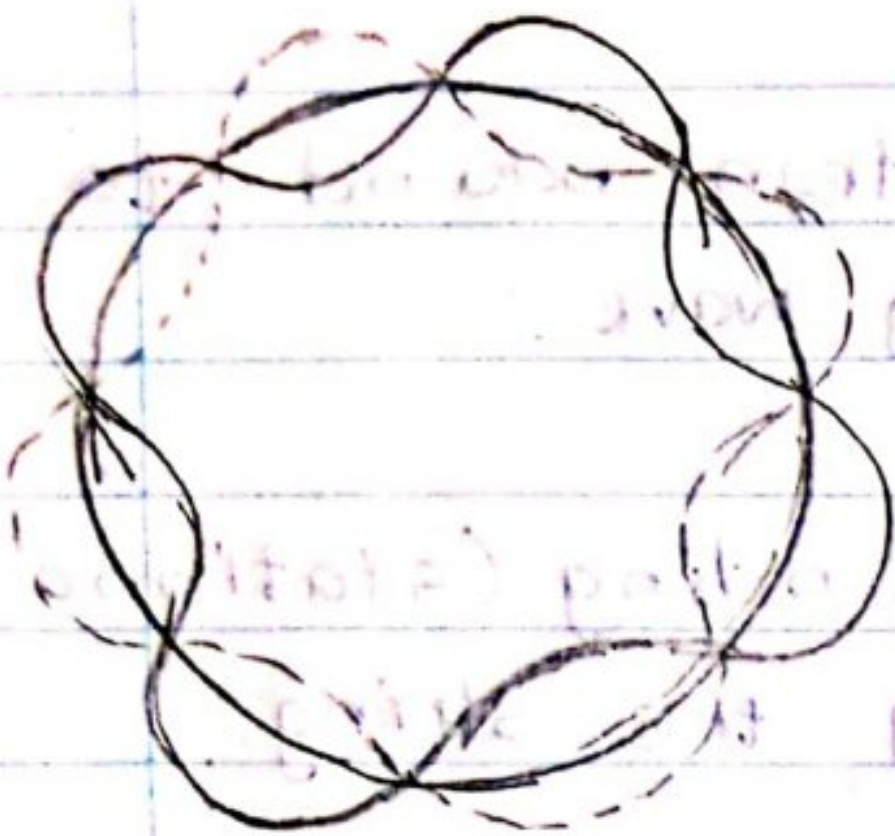


- * De Broglie argued that if an e^- does behave like a standing wave in the hydrogen atom, the length of the wave must fit the circumference of the orbit exactly.



* The circumference of the orbit is not equal to an integral number of wavelengths. (As a result, the electron wave does not close on itself)

non allowed - orbit



* The circumference of the orbit is equal to an integral number of wavelengths.

allowed - orbit

length of the
Bohr orbit

* Should be equal to whole number multiple of the wave length

$$n\lambda = 2\pi r \quad \lambda = \frac{h}{mv}$$

Bohr radius
(radius of the atom)

$$r = \frac{nh}{2\pi mv}$$

* This equation is generated the Bohr's quantization hypothesis.

The Wave function

In quantum mechanics, a wave function is a quantity that is used to describe the displacement of a matter wave produced by particles (i.e. electrons). By taking the square of the absolute value of the wave function, also allows us to determine the probability of finding the particle at some given location.

Normally matter waves are described by a complex-value wave function (ψ) whose absolute square $|\psi|^2 = \psi \psi^*$ gives the probability of finding the particle at a given point at some instant.

ψ^* is the complex conjugate of ψ . For example, if Z is a complex number,

$$Z = x + iy$$
$$|Z|^2 = Z Z^*$$
$$= (x + iy)(x - iy) = x^2 + y^2$$

Similar manner, when $\psi = e^{ikx}$

$$|\psi|^2 = \psi \psi^* = e^{ikx} e^{-ikx} = e^0 = 1$$

Atlas

- Classical theory and quantum theory blends when the De Broglie wavelength is small compared with the dimensions of the system.
- De Broglie wave ψ depends on both the position and time of all the particles in a system and therefore is often written as $\psi(x, y, t)$
- If ψ is known for a particle, then the particular properties of that particle can be described.

De Broglie equation relates the momentum of a particle to its wavelength through the relation $p = h/\lambda$. So, if a free particle has a precisely known momentum, its wave function is a sinusoidal wave of wavelength $\lambda = h/p$.

The real part of the wave function for such particle moving along x -axis can be written in the form,

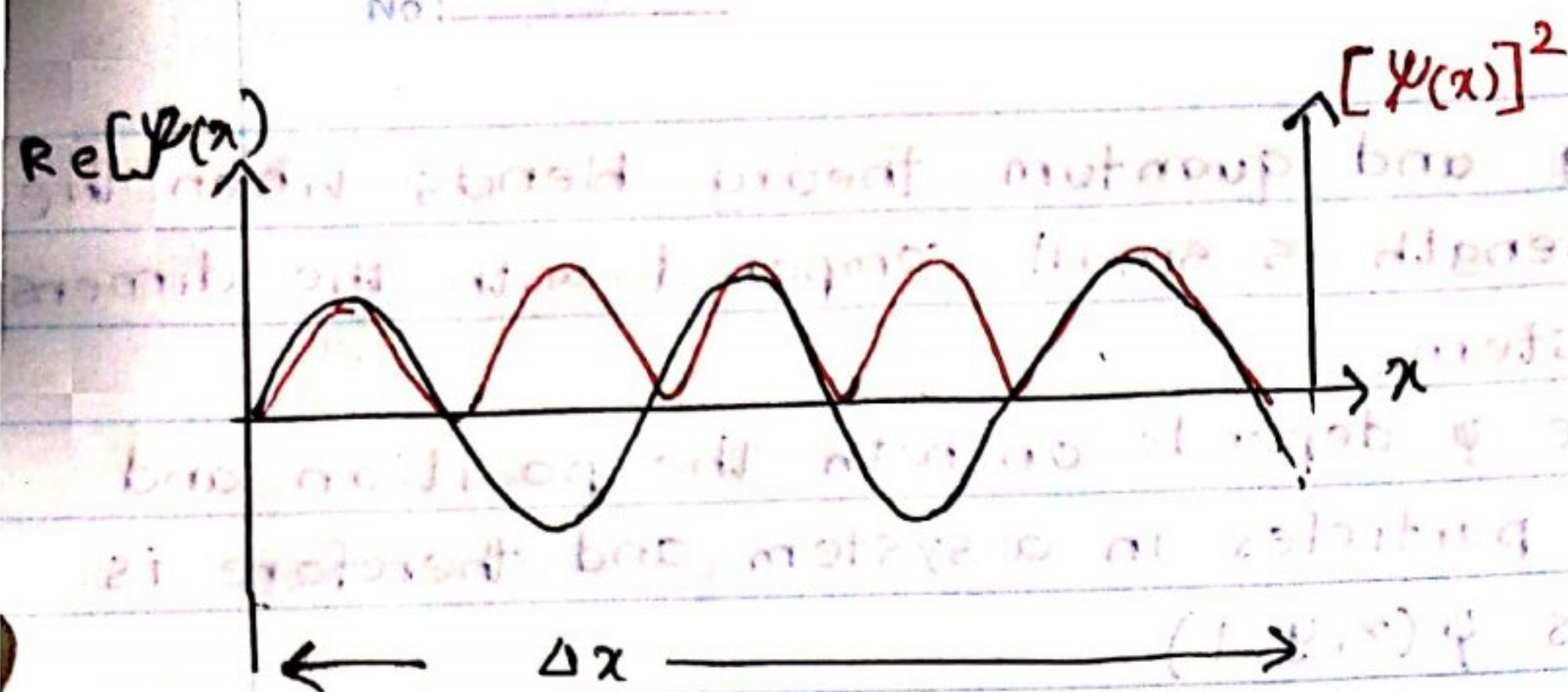
$$\psi(x) = A \sin \left(\frac{2\pi x}{\lambda} \right) = A \sin kx$$

where, $k = \frac{2\pi}{\lambda}$ is the wave number and A is a constant.

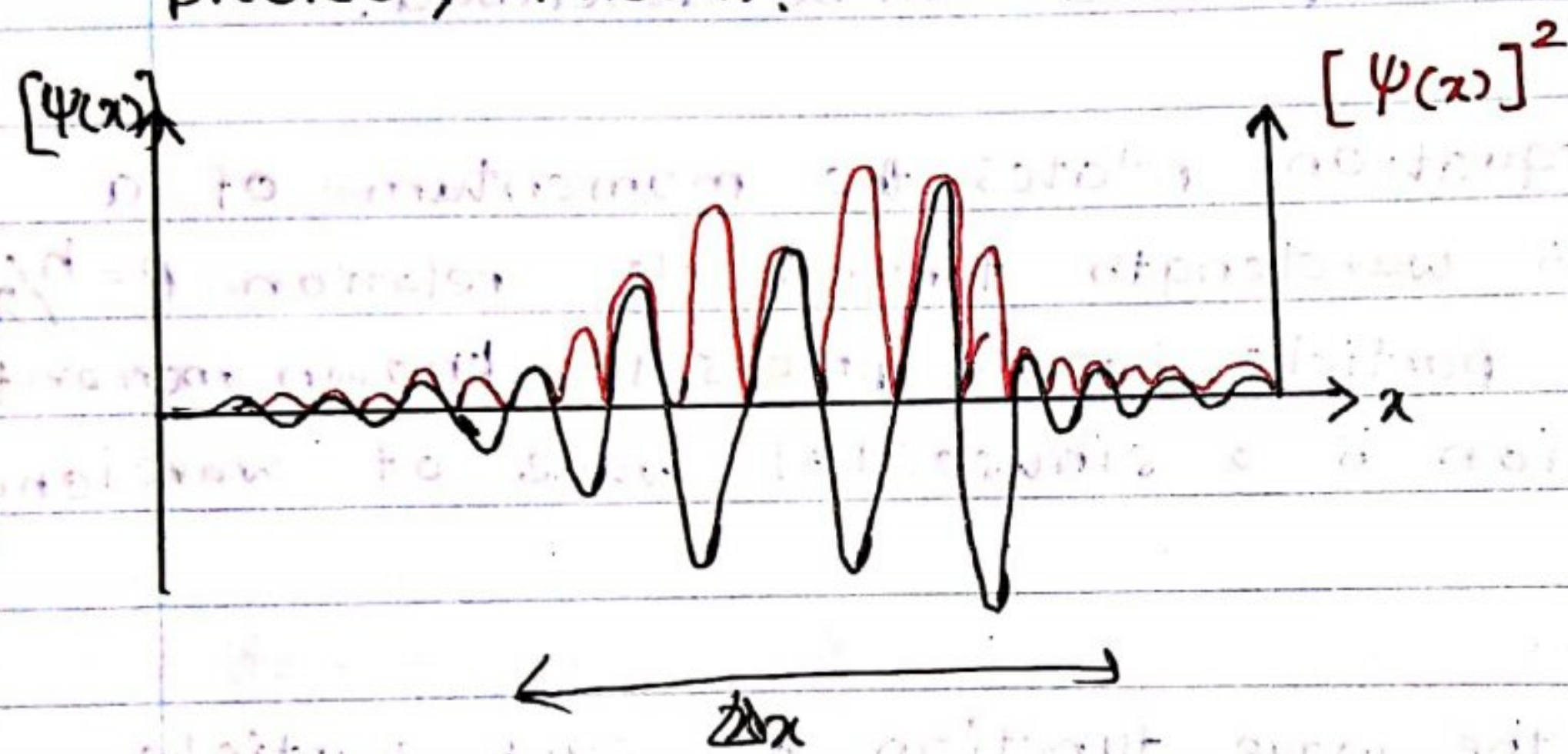
In general, we would write the wave function for the particle in the form Ae^{ikx} , where the imaginary part of the function describes the phase of the wave.

$\psi(x)$ = x dependent wave function

$\psi(\vec{r})$ = x, y, z dependent function



Wave function for a particle whose wavelength is precisely known.



Wave function for a particle whose wavelength isn't precisely known and hence whose momentum is known only over some range of values.

- ψ itself is not a quantity that one can measure.
- $|\psi|^2$ can be measured. If ψ represents a single particle, then $|\psi|^2$ is the probability per unit volume that the particle will be found at any given point. First suggested by Max Born in 1928.

If dv is a small volume element surrounding some point, then the probability of finding the particle in that volume is given by,

$$\text{Probability} = |\psi|^2 dv$$

- Consider a one dimensional system. Then the particle must be located along the x -axis. Therefore the probability that the particle will be found in the infinitesimal interval dx about the point x , denoted by $p(x)dx$ is

$$p(x)dx = |\psi|^2 dx$$

- Since the particle must be somewhere along the x -axis, the sum of the probabilities over all values of x must be 1.

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

Any wave function satisfying the above equation said to be **normalized**. $|\psi|^2$ = probability density.

- Normalization is simply a statement that the particle exists at some point at all times. If the probability were zero, the particle would not exist.

- The probability of finding the particle in the interval $a \leq x \leq b$ is given by

$$P_{ab} = \int_a^b |\psi|^2 dx$$

- The value of the probability must lie between the limits 0 and 1.

eg: If the probability is 0.3, this would signify 30% chance of finding the body.

- Wave function ψ satisfies a wave equation
- The wave equation satisfied by ψ called the Schrödinger equation cannot be derived from any more fundamental laws.

• Although ψ itself is not a measurable quantity, all measurable quantities such as the energy and momentum of a particle can be derived from the knowledge of ψ .

• It is possible to calculate the average position x of the particle. This average position is called the expectation value of x and is defined by

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\psi|^2 dx \quad \text{---} \odot$$

Example :

The normalized wave function of a particle is

$$\psi(x) = \frac{e^{-|x|/x_0}}{\sqrt{x_0}}$$

Find the expectation value of position

$$\langle x \rangle = \int_{-\infty}^{+\infty} x |\psi(x)|^2 dx$$

$$= \int_{-\infty}^{+\infty} x \left| \frac{e^{-|x|/x_0}}{\sqrt{x_0}} \right|^2 dx$$

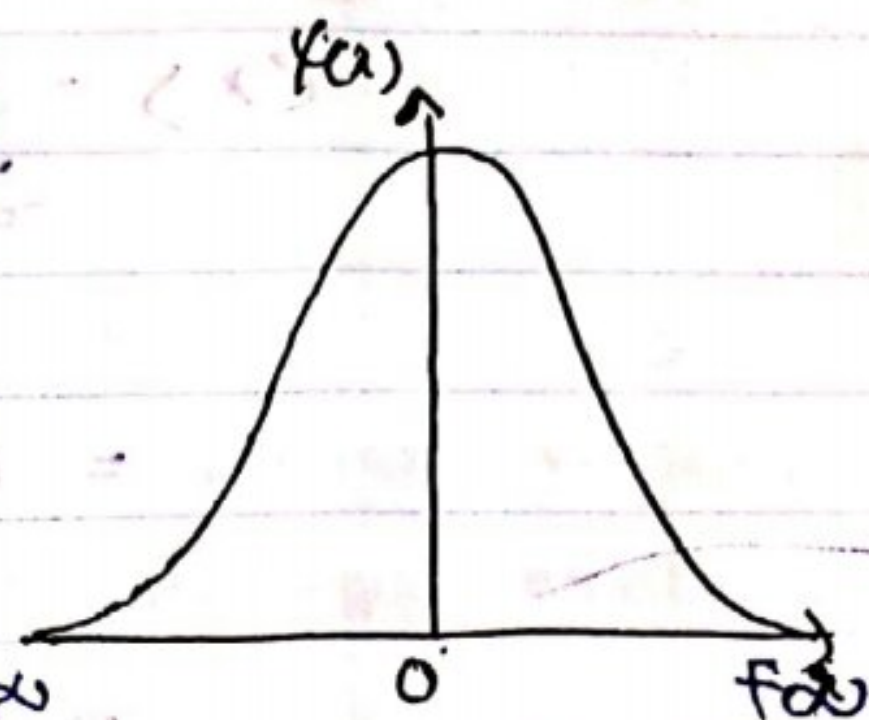
$$= \frac{1}{x_0} \int_{-\infty}^{+\infty} x \cdot e^{-2|x|/x_0} dx$$

$$= \frac{1}{x_0} \cdot 0 = 0$$

No. _____ Date _____

Example 02: Consider a particle whose wave function is graphed and is given by,

$$\psi(x) = Ae^{-ax^2}$$

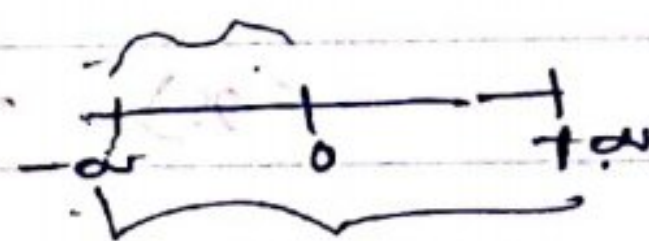


(A) What is the value of A if this wave function is normalized?

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

$$\int_{-\infty}^{\infty} |Ae^{-ax^2}|^2 dx = 1$$

$$A^2 \int_{-\infty}^{\infty} e^{-2ax^2} dx = 1$$



$$A^2 \left(\int_0^{\infty} e^{-2ax^2} dx + \int_{-\infty}^0 e^{-2ax^2} dx \right) = 1$$

$$A^2 \left(- \int_{\infty}^0 e^{-2ax^2} dx (dx) + \int_{-\infty}^0 e^{-2ax^2} dx \right) = 1$$

$$A^2 \left(\int_0^{\infty} e^{-2ax^2} dx + \int_0^{\infty} e^{-2ax^2} dx \right) = 1$$

$$2A^2 \int_0^{\infty} e^{-2ax^2} dx = 1$$

$$\boxed{\int_0^{\infty} e^{-2ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{2a}}}$$

$$\frac{1}{2} A^2 \frac{1}{2} \sqrt{\frac{\pi}{2a}} = 1$$

$$A = \left(\frac{2a}{\pi} \right)^{1/4}$$

Atlas

(B) What is the expectation value of x for this particle?

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* x \psi dx = \int_{-\infty}^{\infty} \{ A e^{-ax^2} \}^2 x dx$$

$$= A^2 \int_{-\infty}^{\infty} x e^{-2ax^2} dx$$

$$= A^2 \left[\int_{-\infty}^0 x e^{-2ax^2} dx + \int_0^{\infty} x e^{-2ax^2} dx \right]$$

$$= A^2 \left[- \int_0^{\infty} x e^{-2ax^2} dx + \int_0^{\infty} x e^{-2ax^2} dx \right]$$

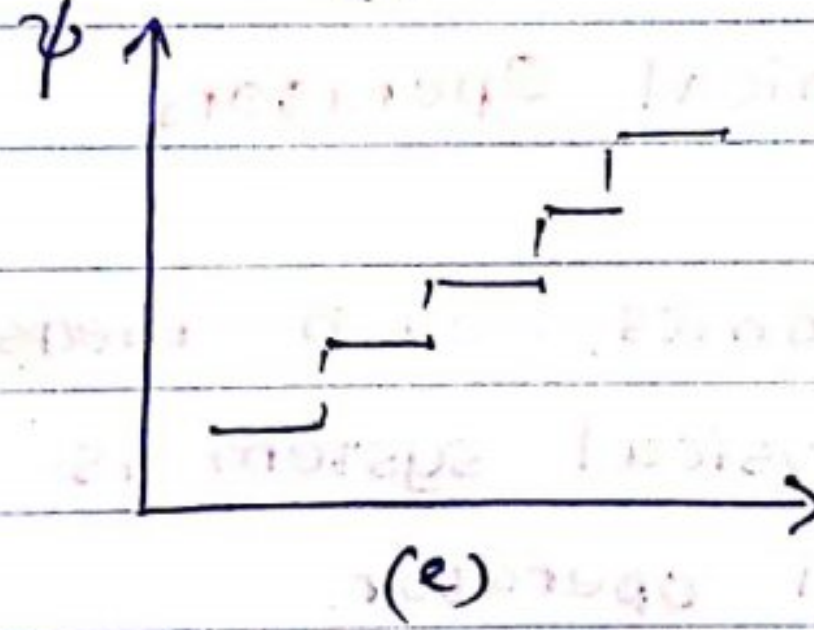
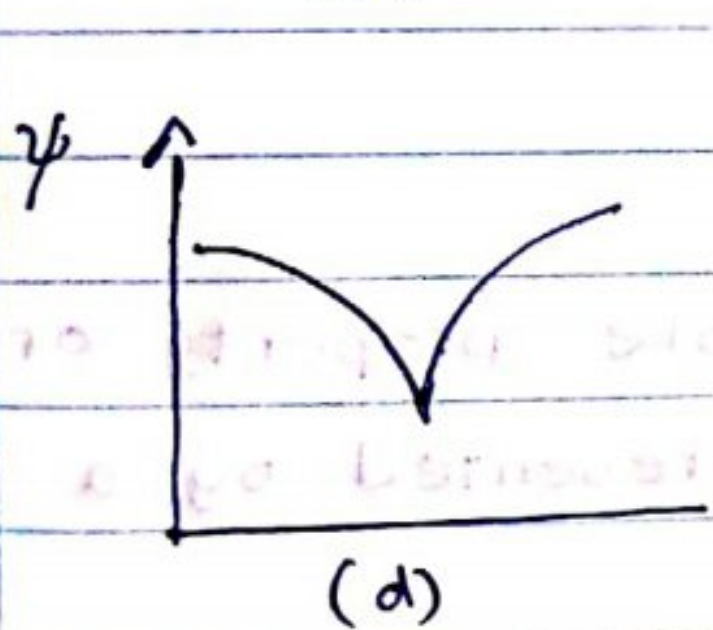
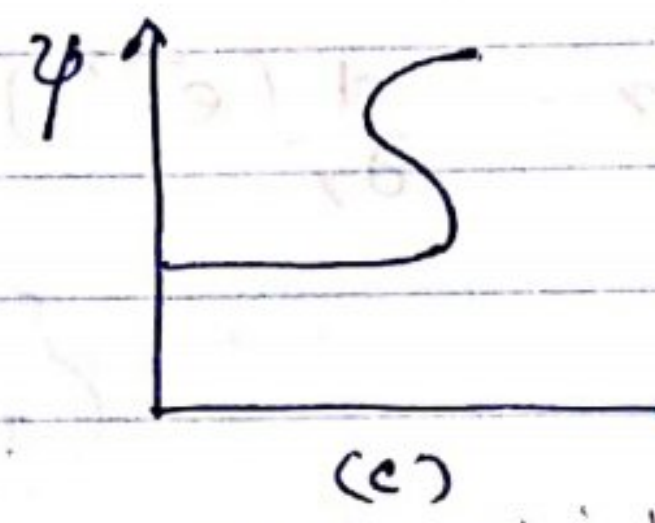
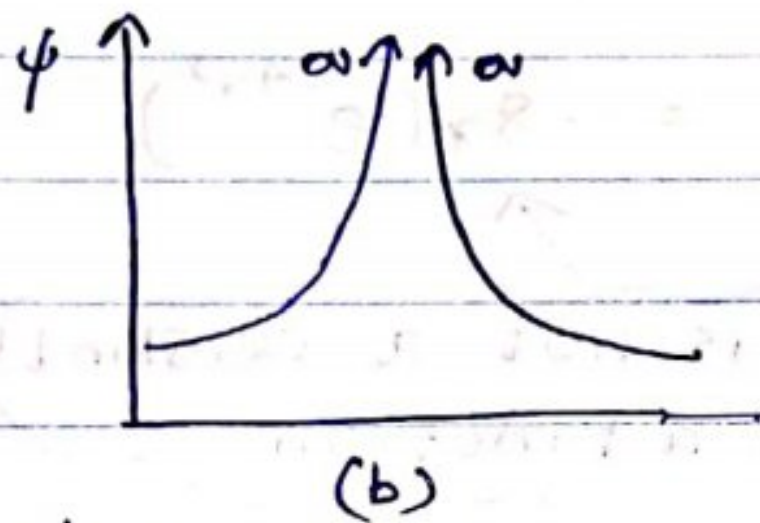
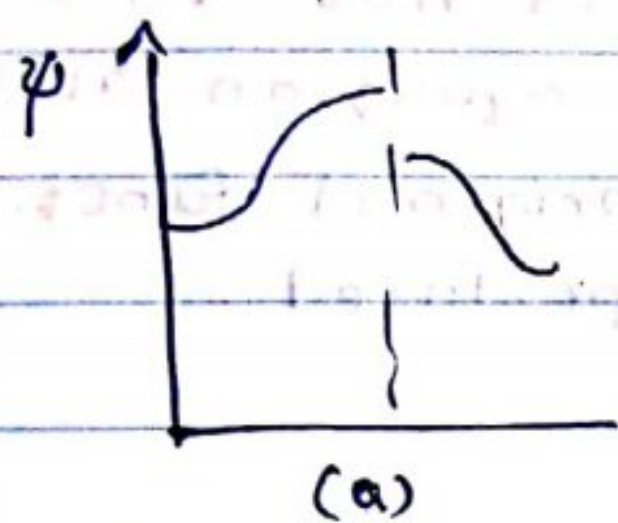
$$\langle x \rangle = 0$$

$$1 = \left[\frac{x e^{-2ax^2}}{-2a} \right]_0^{\infty} + \left[\frac{x e^{-2ax^2}}{-2a} \right]_0^{\infty} = 1$$

Summary of wavefunction

Acceptability of wavefunctions

- (1) - must be single valued everywhere it is allowed to exist
- (2) - must be finite everywhere it is allowed to exist
- (3) The wavefunction and its first derivative must be continuous everywhere
- (4) - span the complex space, ($\psi \in \mathbb{C}$)



not single valued

unacceptable wavefunctions

Born rule: The probability density of finding the particle at a given point is proportional to the square of the magnitude of the particle's wavefunction at that point.

$$|\psi(\vec{r}, t)|^2 = \psi(\vec{r}, t) \psi^*(\vec{r}, t) \leftarrow \text{Probability density}$$

Normalization

$$\int_{-\infty}^{\infty} |\psi(\vec{r}, t)|^2 dV = \int_{-\infty}^{\infty} \psi(\vec{r}, t) \psi^*(\vec{r}, t) dV = 1$$

Eigen value Equation

operator - transform one function to another function

$$\hat{O} f(x) = c(f(x))$$

operator eigenvalue

$$\text{ex: } \frac{d^2}{dx^2} \left(\cos \frac{x}{4} \right) = \left(-\frac{1}{16} \right) \cos \frac{x}{4}$$

← eigen value

$$\text{ex: } \frac{d}{dx} (e^{-4x^2}) = (-8x)(e^{-4x^2})$$

It is not a constant
It is a function

This is not an eigen value equation although the original function is reproduced

Quantum Mechanical Operators

- In quantum mechanics, each measurable property or an observable in a physical system is represented by a quantum mechanical operator.

$$(1) \hat{X} = (x, y, z) = x\hat{x} + y\hat{y} + z\hat{z} \quad (\text{position operator})$$

$$(2) \hat{p} = -i\hbar \vec{\nabla} \quad (\text{momentum operator})$$

$$\therefore \vec{\nabla} = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x} \quad (\text{momentum operator in single dimension acting along } x \text{ axis})$$

* Operators corresponding to other observable, such as velocity, angular momentum, can derive from the position and momentum operators based on the classical relationships exist between these observable

- and position & momentum

ex:- $\vec{v} = \frac{\vec{p}}{m}$ ← momentum
 ← particle of mass

quantum velocity operator $\hat{v}$ is given by

$$\hat{V} = i \frac{\hbar}{m} \vec{\nabla}, \quad (3D)$$

and

$\hat{V}_x = -i \frac{\hbar}{m} \frac{\partial}{\partial x}$, is the component of the velocity in x axis

Expectation values

- average value observed for the measurement

$$\langle A \rangle = \int \psi^* \hat{A} \psi dV$$

$\hat{A}$ is the quantum operator corresponding to the particular classical observable (position, momentum, etc)

Exercise 2.13

- a) Establish the classical relationship between momentum $\vec{p}$ and kinetic energy, K .
- b) Extracting from the above result, derive expressions for the quantum kinetic energy operator $\hat{K}$,
- for system occur in three dimension
 - for a system confined to single dimension (x, y, or z)

(show all steps involved in the derivations).

$$K = \frac{1}{2} m v^2$$

$$K = \frac{1}{2} (mv) v$$

$$K = \frac{1}{2} p v$$

$$\hat{K} = \frac{1}{2} \hat{p} \hat{v}$$

$$= \frac{1}{2} \left(-i \frac{h}{2\pi} \vec{\nabla} \right) \left(-i \frac{h}{2\pi m} \vec{\nabla} \right)$$

$$\hat{K} = \frac{-h^2}{8\pi^2 m} \vec{\nabla}^2 \quad (\text{in 3 dimensions})$$

for a single dimension

$$\hat{K} = \frac{-h^2}{8\pi^2 m} \frac{\partial}{\partial x} \quad \text{for } x$$

$$\hat{K} = \frac{-h^2}{8\pi^2 m} \frac{\partial}{\partial y} \quad \text{for } y$$

$$\hat{K} = \frac{-h^2}{8\pi^2 m} \frac{\partial}{\partial z} \quad \text{for } z$$

- a) Provide the classical definition for angular momentum, $\vec{L}$ that involve momentum $\vec{p}$.
- b) Using the above relationship, derive an expressions for the quantum angular momentum operator $\hat{L}$,

i. For a three dimensional system. $\hat{L} = L_x + L_y + L_z$

- ii. Extracting from the above result, separately present the x, y, z, components of the angular momentum operator ($\hat{L}_x$, $\hat{L}_y$, and, $\hat{L}_z$)

(show all steps involved in the derivations).

(a) $\vec{L} = \vec{r} \times \vec{p}$: (classical definition of the angular momentum)

where, $\vec{r}$ = position
 $\vec{p}$ = momentum

(b) i) In terms of cartesian unit vectors $\{\hat{x}, \hat{y}, \hat{z}\}$

$\vec{r}$ and $\vec{p}$ take the form

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\vec{p} = p_x\hat{x} + p_y\hat{y} + p_z\hat{z}$$

$$\vec{L} = \vec{r} \times \vec{p} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

$$\Rightarrow \hat{L} = \hat{x}(y p_z - z p_y) - \hat{y}(x p_z - z p_x) + \hat{z}(x p_y - y p_x)$$

$$\text{is } L_x = y p_z - z p_y$$

$$L_y = z p_x - x p_z$$

$$L_z = x p_y - y p_x$$

The quantum angular momentum operator $\hat{L}$ is given by

$$\hat{L} = \hat{x}(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y) + \hat{y}(\hat{z}\hat{p}_x - \hat{x}\hat{p}_z) + \hat{z}(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)$$

Where, $\hat{P}_x, \hat{P}_y, \hat{P}_z$ are the quantum momentum operators in 1D, such that

$$\hat{P}_x = -i\hbar \frac{\partial}{\partial x} \quad \hat{P}_z = -i\hbar \frac{\partial}{\partial z}$$

$$\hat{P}_y = -i\hbar \frac{\partial}{\partial y}$$

$$\Rightarrow \hat{L} = \hat{r} \times \hat{p}$$

$$(ii) \quad \hat{L}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right)$$

$$\hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

$$\hat{L}_y = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right)$$

Classical Mechanics:
Equation of Motion
'Newton's 2nd Law ($F=ma$)'

Quantum Mechanics
Equation of Motion
'Schrodinger Equation'

Schrodinger Equation - explains the time evolution of the wave function.

$$i\hbar \frac{d\psi}{dt} = \hat{H}\psi$$

$\hat{H}$ = quantum Hamiltonian operator'

$$\hat{H} = \hat{K} \cdot E + \hat{P} \cdot E$$

↑ Kinetic energy operator ↓ Potential energy operator

Schrodinger Equation

Time-Dependent

$(\hat{x}, \hat{p}, t)$

Time-Independent

$(\hat{x}, \hat{p})$ (point at stationary state)

For a Free particle

(Free particle - there is no force acting on it)

$$F = -\frac{\partial V(r)}{\partial r} \quad \therefore V_r = \text{constant}/0$$

$$E_{\text{total}} = \frac{1}{2}mv_{(x)}^2 + \underbrace{V_{(x)}}_0$$

$$E_{\text{total}} = \frac{\hat{p}^2}{2m}$$

$$i\hbar \frac{\partial(\psi(x,t))}{\partial t} = \frac{-\hbar^2}{2m} \frac{\partial^2(\psi(x,t))}{\partial x^2}$$

Particle moving in a Potential

$$E_{\text{total}} = K.E + V$$

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi(\vec{r}, t)$$

$$\psi(\vec{r}, t) = \phi(\vec{r}) f(t) \leftarrow \text{function of time only}$$

In time-independent Schrödinger equation we consider only function of position,

$$E \phi(\vec{r}) = \left(-\frac{\hbar^2}{2m} \nabla^2 + \hat{V}(\vec{r}) \right) \phi(\vec{r})$$

$$\therefore \nabla^2 = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

Time-independent Schrödinger equation from time-dependent equation

For one dimension,

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + \hat{V}(x) \psi(x, t) \right)$$

Since the potential of time independent the wave function can be considered as a product function of two separate functions.

$$\psi(x, t) = \phi(x) f(t)$$

Substitute this in the time-dependent Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} (\phi(x) f(t)) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (\phi(x) f(t)) + V(x) (\phi(x) f(t))$$

$$i\hbar \phi(x) \frac{\partial}{\partial t} (f(t)) = -\frac{\hbar^2}{2m} f(t) \frac{\partial^2}{\partial x^2} (\phi(x)) + V(x) \phi(x) f(t)$$

* $\div f(t)$

$$\phi(x) \left[\frac{i\hbar \frac{\partial}{\partial t} (f(t))}{f(t)} \right] = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (\phi(x)) + V(x) \phi(x)$$



E - time dependant

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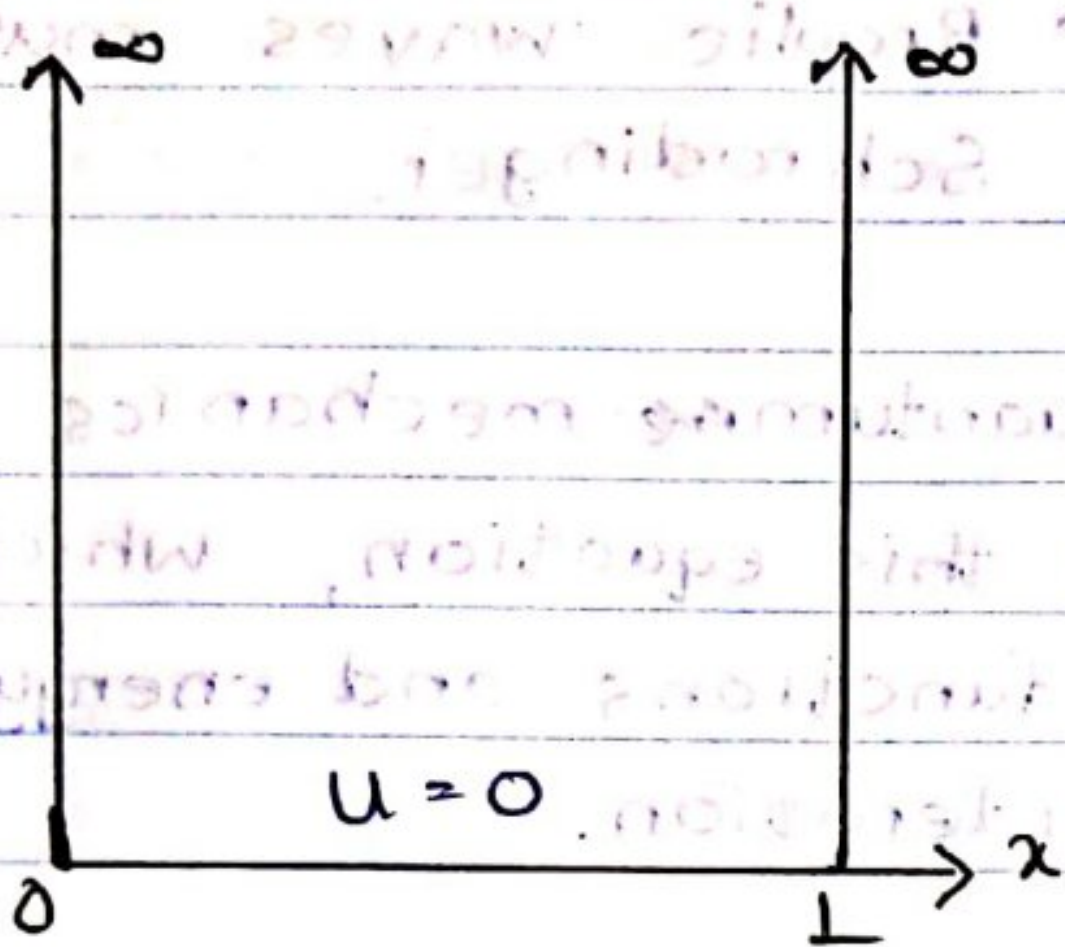
$$\frac{\partial}{\partial t} f(t) = -i \frac{E}{\hbar} f(t)$$

It shows that E is an eigenvalue of the Hamiltonian operator corresponding to the eigenstate $\phi(x)$.

$$E \phi(x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \phi(x) + V(x) \phi(x)$$

The particle in a box
(Particle in an infinite potential well)

(a)



The walls are impenetrable, there is zero probability of finding the particle outside the box, so the wave function $\psi(x)$ must be zero for $x < 0$ and $x > L$.

And it must be zero at the walls, that is $\psi(0) = 0$ and $\psi(L) = 0$.

The walls are infinitely high.

So, $U(x) = \infty$ for $x = 0$ & $x = L$

$U(x) = \infty$ for $x = L$

The potential energy is constant within the box, and is convenient to choose $U = 0$ in this region for $0 < x < L$, $U(x) = 0$.

Schrödinger equation

$$\left[\frac{-\hbar^2}{8\pi^2 m} \frac{d^2 \psi}{dx^2} + U(x) \right] \psi = E \psi$$

$$\frac{-\hbar^2}{8\pi^2 m} \frac{d^2 \psi}{dx^2} = E \psi$$

$$\frac{d^2 \psi}{dx^2} = -\frac{8\pi^2 m E}{\hbar^2} \psi = -k^2 \psi \quad \text{--- (1)}$$

$$k = \frac{\sqrt{8\pi^2 m E}}{\hbar}$$

$$\frac{d^2 \psi}{dx^2} = -k^2 \psi$$

We expect oscillating solutions, so the most general solution to this equation is

$$\psi(x) = A \sin Kx + B \cos Kx$$

Now, $\psi(x) = 0$ at $x = 0$

$$\psi(0) = A \sin K \cdot 0 + B \cos(K \cdot 0)$$

$$0 = B$$

$$\psi(x) = 0 \text{ at } x = L$$

$$\psi(L) = A \sin K \cdot L + \underbrace{B \cos K \cdot L}_0$$

$$A \sin KL = 0$$

$$A \neq 0 \text{ or } \sin KL = 0$$

$$KL = n\pi \quad n \text{ is an integer}$$

$$K = \frac{\sqrt{8mEL^2E}}{h}$$

$$\frac{n\pi}{L} = \frac{\sqrt{8m\pi^2E}}{h} \Rightarrow \frac{n\pi h}{L} = 2\sqrt{2} \sqrt{mE} \pi$$

$$\frac{nh}{2L} = \sqrt{E}$$

$$\therefore E = \frac{n^2 h^2}{8mL^2} \quad n = \text{integer}$$

Allowed wave functions are given by

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{L}\right)$$

From normalization,

$$\int_0^L A \sin\left(\frac{n\pi x}{L}\right) A \sin\left(\frac{n\pi x}{L}\right) dx = 1$$

$$A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

$$A = \sqrt{\frac{2}{L}}$$

$n=1$;

$$E_1 = \frac{h^2}{8mL^2}$$

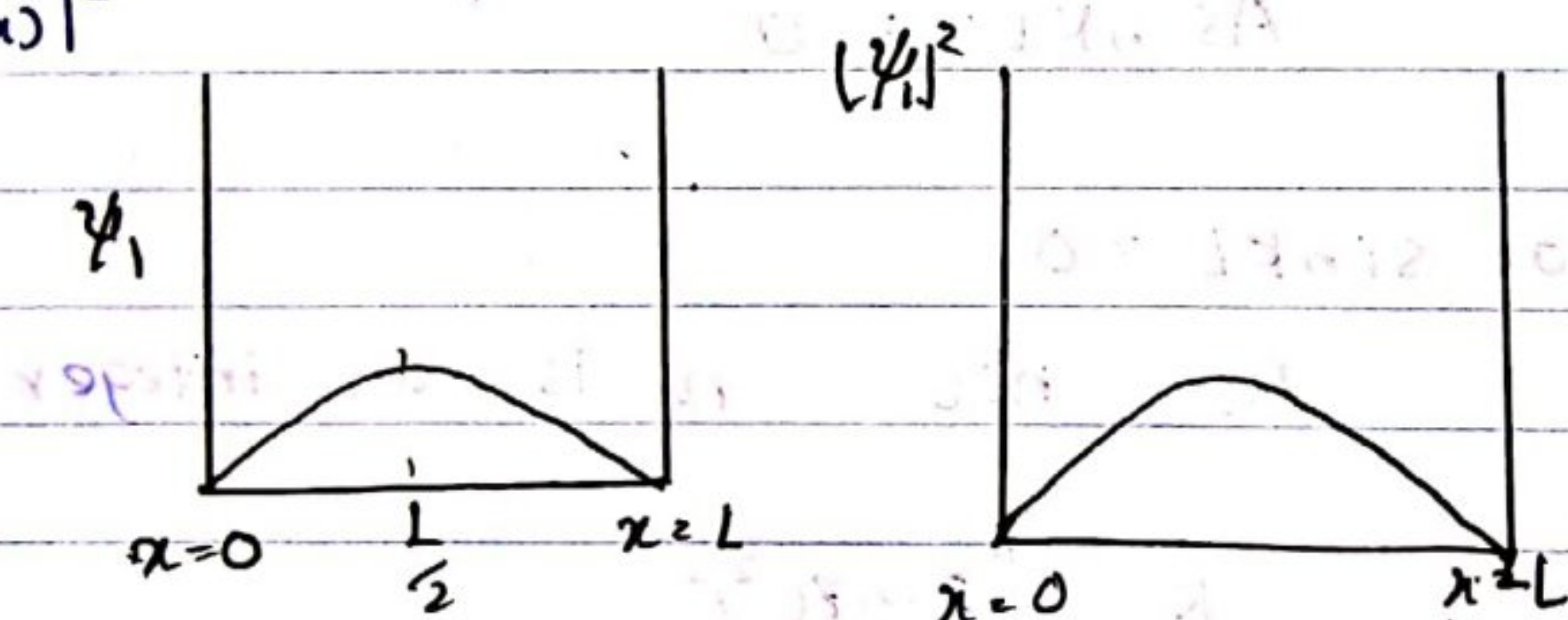
ground state $\psi_1(x) = A \sin\left(\frac{\pi x}{L}\right)$

$n=2$;

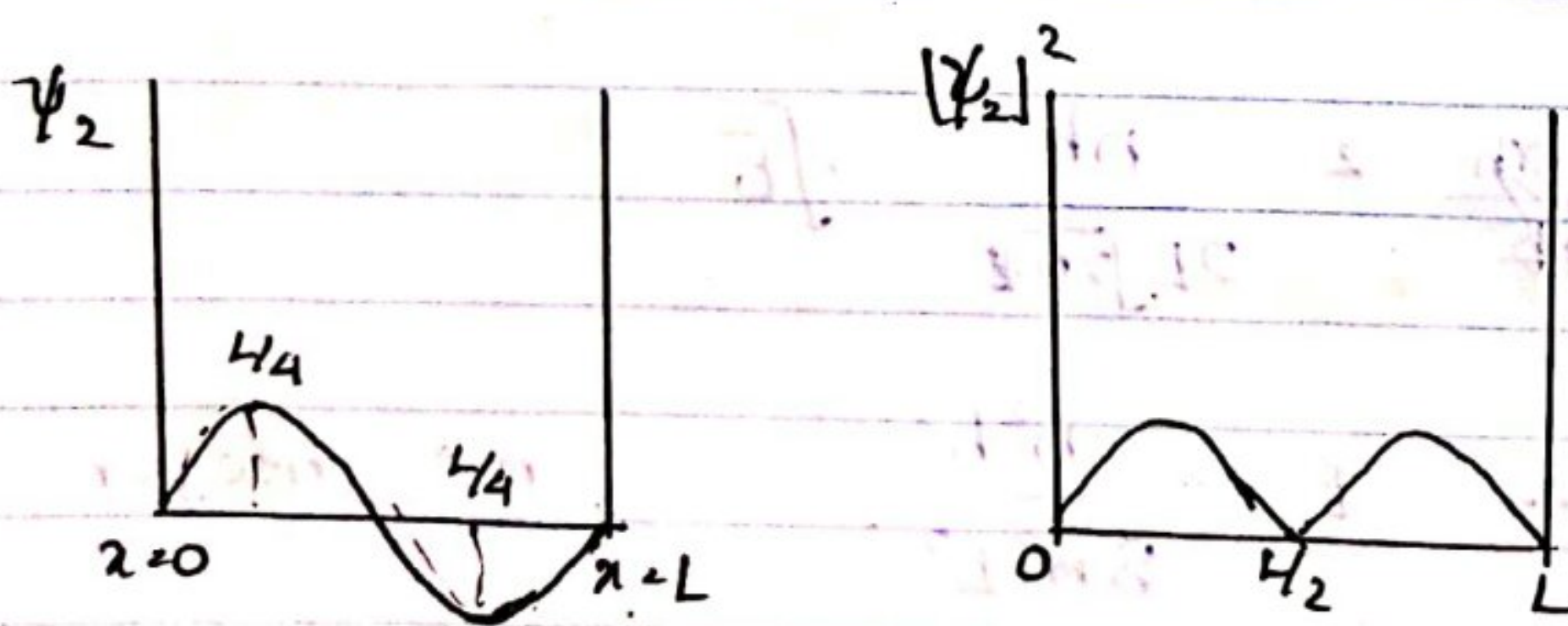
$$E_2 = \frac{4h^2}{8mL^2}$$

1st excited state $\psi_2(x) = A \sin\left(\frac{2\pi x}{L}\right)$

1) Ground state, $\psi_1(x) = A \sin\left(\frac{\pi x}{L}\right)$ and probability density $P = |\psi_1(x)|^2$



2) 1st excited state, $\psi_2(x) = A \sin\left(\frac{2\pi x}{L}\right)$ and probability density $P = |\psi_2(x)|^2$



PHY 2106

Atomic and Nuclear Physics

1-D time independent Schrödinger Equation:

$$\underbrace{\frac{-\hbar^2}{2m} \frac{d^2\Psi}{dx^2}}_{\text{KE}} + \underbrace{U\Psi}_{\text{PE}} = \underbrace{E\Psi}_{\text{Total E}}$$

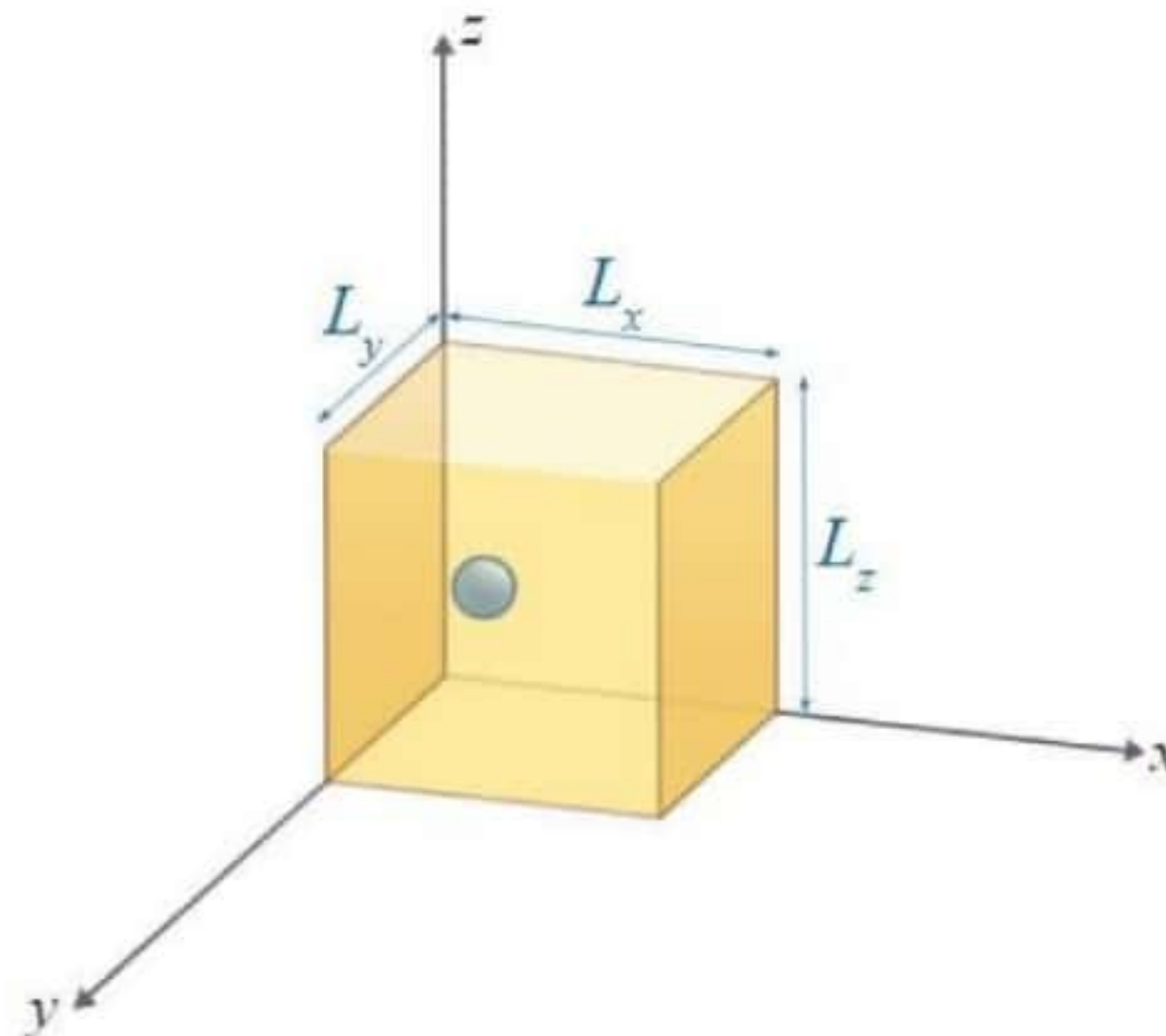
m - mass of the particle

U - potential energy of the system

E - total energy of the system

Particle in a 3-D box:

The quantum particle in the 1D box problem can be expanded to consider a particle within a higher dimension as a particle in a 2D box or 3D box. Here we continue the expansion into a particle trapped in a 3D box with three lengths L_x , L_y , and L_z . As with the other systems, there is no force (i.e., no potential) acting on the particles inside the box.



The potential for the particle inside the box $V(\vec{r}) = 0$

$\vec{r}$ is the vector with all three components along the three axes of the 3-D box:

$$\vec{r} = L_x \hat{x} + L_y \hat{y} + L_z \hat{z}$$

3-D Time independent Schrodinger equation:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(r) + V(r) \psi(r) = E \psi(r)$$

$$-\frac{\hbar^2}{2m} \left(\frac{d^2 \psi(r)}{dx^2} + \frac{d^2 \psi(r)}{dy^2} + \frac{d^2 \psi(r)}{dz^2} \right) = E \psi(r)$$

To solve this equation, take the wave function equal to a product of individual function for each independent variable. (using the separation of variables technique)

$$\psi(x, y, z) = X(x)Y(y)Z(z)$$

Now each function has its own variable:

- $X(x)$ is a function of variable x only
- $Y(y)$ is a function of variable y only
- $Z(z)$ is a function of variable z only

Now substitute in above equation and divide by XYZ product:

$$\frac{d^2 \psi}{dx^2} = YZ \frac{d^2 X}{dx^2} \Rightarrow \frac{1}{X} \frac{d^2 X}{dx^2}$$

$$\frac{d^2 \psi}{dy^2} = XZ \frac{d^2 Y}{dy^2} \Rightarrow \frac{1}{Y} \frac{d^2 Y}{dy^2}$$

$$\frac{d^2 \psi}{dz^2} = XY \frac{d^2 Z}{dz^2} \Rightarrow \frac{1}{Z} \frac{d^2 Z}{dz^2}$$

We can rearrange the equation;

$$\left(-\frac{\hbar^2}{2mX} \frac{d^2 X}{dx^2} \right) + \left(-\frac{\hbar^2}{2mY} \frac{d^2 Y}{dy^2} \right) + \left(-\frac{\hbar^2}{2mZ} \frac{d^2 Z}{dz^2} \right) = E$$

Normalized wave function equation in a 3D box;

$$X(x) = \sqrt{\frac{2}{L_x}} \sin\left(\frac{n_x \pi x}{L_x}\right)$$

$$Y(y) = \sqrt{\frac{2}{L_y}} \sin\left(\frac{n_y \pi y}{L_y}\right)$$

$$Z(z) = \sqrt{\frac{2}{L_z}} \sin\left(\frac{n_z \pi z}{L_z}\right)$$

Here, $n_x, n_y, n_z = 1, 2, 3, \dots, \infty$

Therefore, wave function inside a 3D box,

$$\psi(r) = \sqrt{\frac{8}{V}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right)$$

Here,

$$V = \underbrace{L_x \times L_y \times L_z}_{\text{volume of box}}$$

Total energy in 3D box;

$$E_{n_x, n_y, n_z} = \frac{h^2}{8m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right)$$

Degeneracy in a 3D cube:

The energy of the particle in a 3-D cube in the ground state is given by when, $n_x=1$, $n_y=1$, and $n_z=1$. This energy ($E_{1,1,1}$) is hence,

$$E_{1,1,1} = \frac{3h^2}{8mL^2}$$

The ground state has only one wavefunction and no other state has this specific energy; the ground state and the energy level are said to be **non-degenerate**. However, in the 3-D cubical box potential the energy of a state depends on the sum of the squares of the quantum numbers.

$$n^2 = n_x^2 + n_y^2 + n_z^2$$

The particle having a particular value of energy in the excited state may have several different stationary states or wavefunctions. If so, these states and energy eigenvalues are said to be **degenerate**.

For the first excited state, three combinations of the quantum numbers (n_x, n_y, n_z) are $(2,1,1), (1,2,1), (1,1,2)$. The sum of squares of the quantum numbers in each combination is same (equal to 6). Each wavefunction has same energy:

$$E_{2,1,1} = E_{1,2,1} = E_{1,1,2} = \frac{6h^2}{8mL^2}$$

Corresponding to these combinations three different wavefunctions and **three** different states are possible. Hence, the first excited state is said to be three-fold or triply degenerate. The number of independent wavefunctions for the stationary states of an energy level is called as the **degree of degeneracy** of the energy level.

Degeneracy properties of the particle in a 3D cube with $L_x=L_y=L_z=L$

$n_x^2 + n_y^2 + n_z^2$	Combinations of Degeneracy (n_x, n_y, n_z)	Total Energy (E_{n_x, n_y, n_z})	Degree of Degeneracy
3	(1,1,1)	$\frac{3h^2}{8mL^2}$	1
6	(2,1,1)(1,2,1)(1,1,2)	$\frac{6h^2}{8mL^2}$	3
9	(2,2,1)(1,2,2)(2,1,2)	$\frac{9h^2}{8mL^2}$	3
11	(3,1,1)(1,3,1)(1,1,3)	$\frac{11h^2}{8mL^2}$	3
12	(2,2,2)	$\frac{12h^2}{8mL^2}$	1
14	(3,2,1)(3,1,2)(2,3,1) (2,1,3)(1,3,2)(1,2,3)	$\frac{14h^2}{8mL^2}$	6
17	(2,2,3)(3,2,2)(2,3,2)	$\frac{17h^2}{8mL^2}$	3